FACULTY OF ENGINEERING AND TECHNOLOGY

LAB EXERCISE - 2

Program: CSE * C / GENERAL - E	Semester: IV
Course Name: WEB TECHNOLOGIES	Course Code: 23CSE404
Submission Date: 02/02/2026	Year: II
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Experiment 1: Personal Portfolio Website Construction

1. Objective

To design and implement a structured personal portfolio website using HTML5 to demonstrate proficiency in document linking, semantic structure, and data organization via tables and lists.

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2. Technical Theory

This experiment covers the transition from traditional multi-page navigation to Single Page Application (SPA) architecture:

- **Hyperlinks (<a>): Used to connect multiple independent HTML files (index.html, about.html, etc.).**
- **Internal Linking: Using the id attribute and href fragments (e.g., href="#projects") to navigate within the same document.**
- **Tables (<table>): A structured way to display project data using rows (<tr>) and headers (<th>).**
- **Forms (<form>): Used to collect user input, demonstrating basic interactive elements like <input> and <textarea>.**

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3. **Implementation Steps**

    1. **Multi-Page Setup: Created four separate HTML files (index.html, about.html, projects.html, and contact.html) with a consistent navigation bar to allow cross-page browsing.**
    2. **Information Architecture:**
        - **Developed an "About Me" section using unordered lists (<ul>) to highlight skills.**
        - **Constructed a "Project Gallery" using a table to track assignment statuses.**

    3. **SPA Consolidation: Merged all four pages into a single file (1.html) using <section> tags with unique IDs.**

    4. **Navigation Logic: Updated the navigation menu to use hash-links (e.g., #contact), allowing the browser to jump to specific sections without reloading the page.**

    5. **Interactive Contact Form: Implemented a form section with name, email, and message fields to simulate a real-world "Contact Us" feature.**

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Personal Website - All-in-One</title>
</head>
<body>

  <nav>
    <a href="#home">Home</a> |
    <a href="#about">About Me</a> |
    <a href="#projects">Projects</a> |
    <a href="#contact">Contact</a>
  </nav>
  <hr>

  <section id="home">
    <h1>Welcome to My Website</h1>
```

```
        <p>This is the <b>Home Section</b>. Everything you see here is contained within a single HTML
document.</p>
        <img src="https://via.placeholder.com/300x150" alt="Home Banner">
    </section>

    <br><br><hr><br><br>

    <section id="about">
        <h2>About Me</h2>
        <p>I am a student learning to combine multiple web pages into one single-page application (SPA) layout.</p>
        <h3>My Skills:</h3>
        <ul>
            <li>HTML Layouts</li>
            <li>Internal Page Linking</li>
            <li>Form Creation</li>
        </ul>
    </section>

    <br><br><hr><br><br>

    <section id="projects">
        <h2>My Project Gallery</h2>
        <p>Below is the status of my current HTML assignments:</p>
        <table border="1" cellpadding="10">
            <tr>
                <th>Project Number</th>
                <th>Task Description</th>
                <th>Status</th>
            </tr>
            <tr>
                <td>Project 1</td>
                <td>4-Page Personal Website</td>
                <td>Completed</td>
            </tr>
            <tr>
                <td>Project 2</td>
                <td>Pending...</td>
                <td>Waiting</td>
            </tr>
        </table>
    </section>

    <br><br><hr><br><br>

    <section id="contact">
        <h2>Contact Me</h2>
        <form>
            <label>Full Name:</label><br>
            <input type="text" name="user_name" placeholder="Your Name"><br><br>
```

```
            <label>Email Address:</label><br>
            <input type="email" name="user_email" placeholder="email@example.com"><br><br>

            <label>Your Message:</label><br>
            <textarea rows="4" cols="40"></textarea><br><br>

            <input type="submit" value="Submit Info">
        </form>
    </section>

    <br><br>
    <footer>
        <p align="center">&copy; 2024 Created for HTML Project Assignment</p>
    </footer>

</body>
</html>
```

CSS:

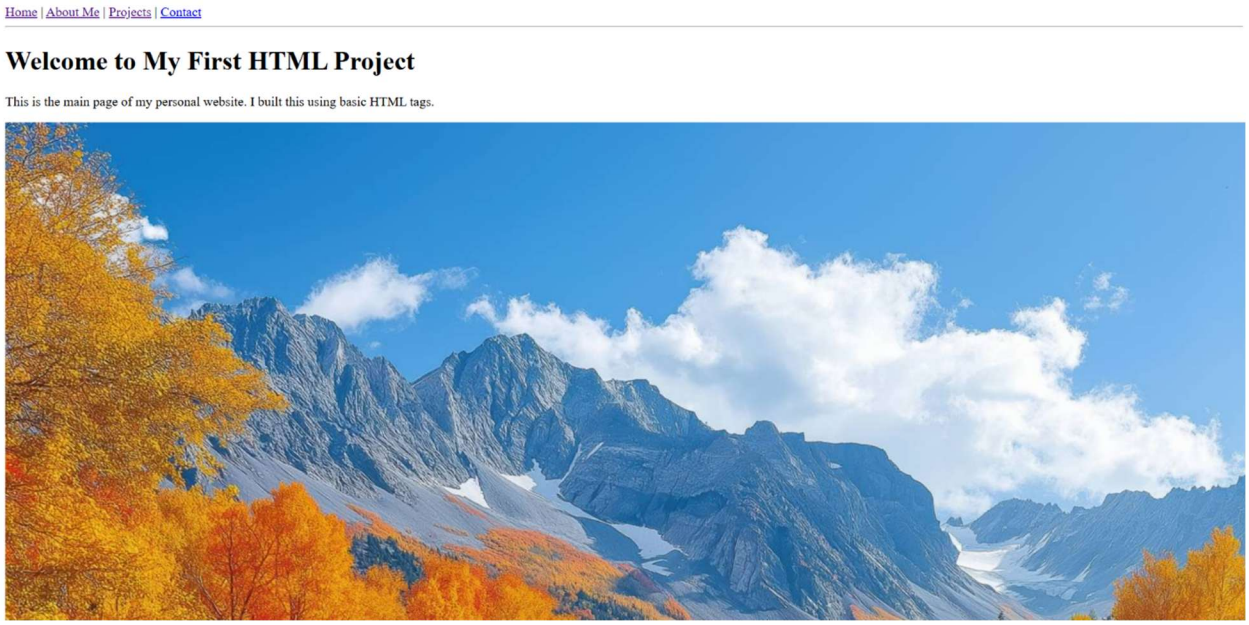
```
/* Calculator Styling */
.calc-container {
    background: #1a1a2e;
    padding: 20px;
    border-radius: 15px;
    width: 320px;
    margin: 20px auto;
    box-shadow: 0 10px 30px rgba(0,0,0,0.5);
}

#display {
    width: 100%;
    height: 50px;
    background: #0f172a;
    color: #00d2ff;
    border: none;
    text-align: right;
    padding: 10px;
    font-size: 24px;
    border-radius: 5px;
    margin-bottom: 15px;
    box-sizing: border-box;
}

.calc-buttons {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 10px;
}
```

```
.calc-btn {
   padding: 15px;
   background: #37475a;
   color: white;
   border: none;
   border-radius: 8px;
   cursor: pointer;
   font-weight: bold;
   transition: 0.2s;
}

.calc-btn:hover { background: #4a5a6e; }

.operator { background: #f39c12; }
.scientific { background: #3498db; }
.equals { background: #27ae60; grid-column: span 2; }
.clear { background: #e74c3c; grid-column: span 2; }
```

5. Output Analysis

- **Structural Integrity**: The use of <section> tags clearly separates the user's bio, projects, and contact information.
- **Navigation Efficiency**: The SPA layout provides a faster user experience as the transitions between "Home" and "Projects" occur instantly.
- **Form Functionality**: The contact section demonstrates how users can interact with the site, providing a clear interface for data entry.

INTIAL OUTPUT WITH HTML



Project List

Project #	Description
01	Personal Website (Current)
02	Waiting for next question...

About Me

I am a student learning web development. Here are my interests:

- Coding
- Design
- Learning New Tech

Contact Me

Your Name:

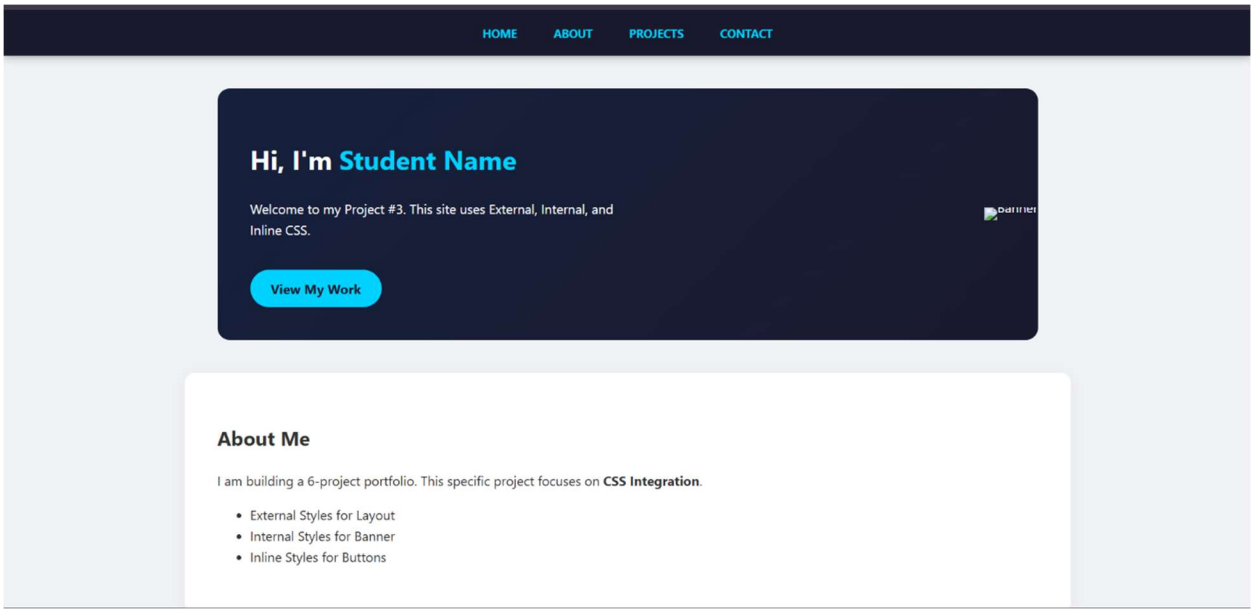
Enter Name

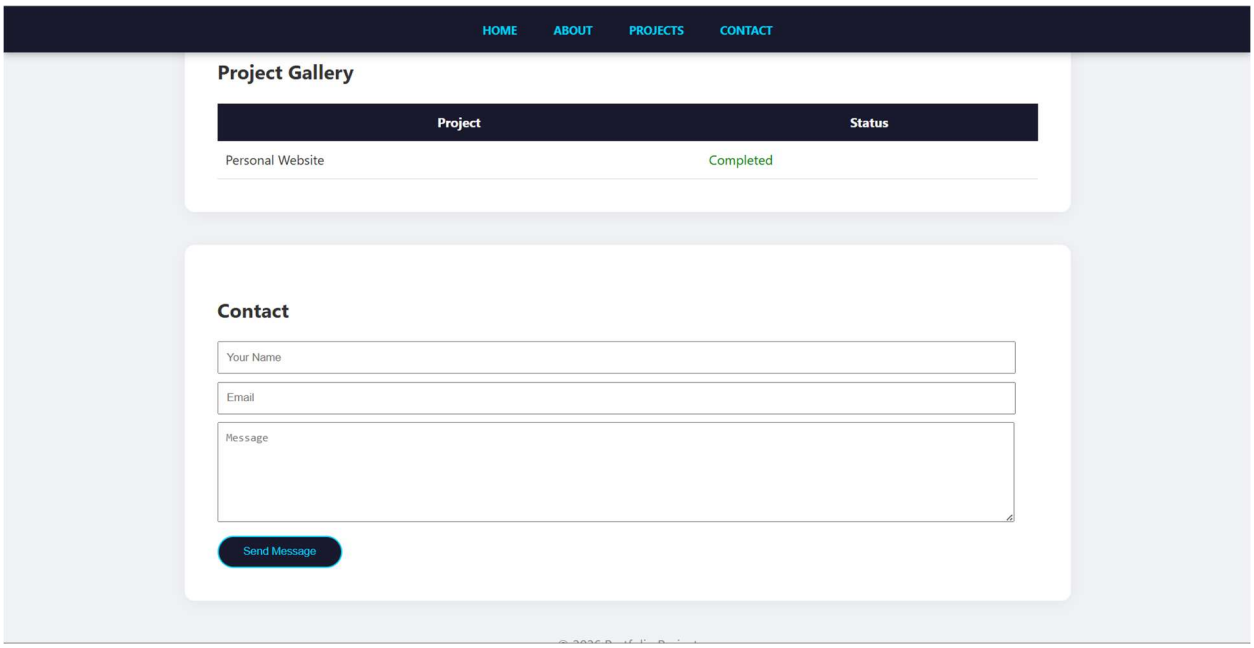
Message:

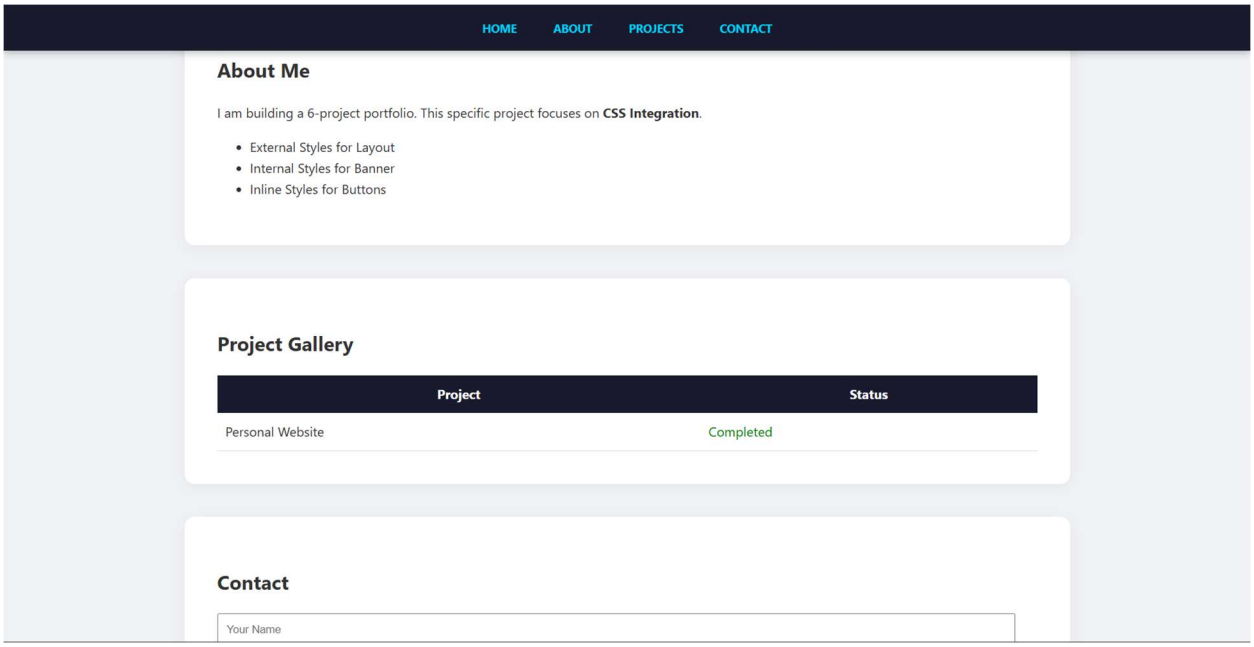
Write something...

Send Message

AFTER IMPLIMENTING CSS:







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6. Conclusion

Experiment 1 successfully demonstrated the fundamentals of HTML organization. By moving from a multi-page structure to an SPA layout, the project highlighted how **internal fragment identifiers** can be used to create efficient, fast-loading personal portfolios. This laid the structural groundwork for more advanced styling and logic in future experiments.